COMP 10280
Programming I (Conversion)

Practical Sheet 16

Déardaoin, 15 Deireadh Fómhair 2026

For each of the following questions, write an algorithm in pseudocode first before writing a Python program. Submit your algorithms in psuedocode as well as your Python programs.
When writing functions, use one-line or multi-line docstrings, as appropriate, to document your functions.

1. Consider the following series, defined recursively as follows:

$$f(n) = \begin{cases} 2 & n = 1 \\ 2 \times f(n-1) & n > 1 \end{cases}$$

   (a) Write a recursive function that takes as its single argument an integer ≥ 1 and returns as a result that term from the above series.

   (b) Write a program that prompts the user for a series of integers. For each number entered the program should check that it is greater than or equal to 1. If it is, it calls the function defined in part (a) and prints out the result. The program should stop when a zero or a negative number is entered.

   (c) In your function, include some print statements that allow you to see the operation of the recursion and its progress towards the base case.

   Save this program as `p16p1.py`.

2. Write a program that prompts the user for a positive integer (≥ 1) and, using the function developed in Question 1(a), prints out that number of terms of the series defined in Question 1. (If you had included `print` statements when developing your function (Question 1(c)), you can discard them for this question.)

   Save this program as `p16p2.py`.

3. Implement the program to calculate common divisors from Lecture 16. You may use the solution on Pages 14–17 of the slides in your solution.

   Save this program as `p16p3.py`.

4. A *perfect number* is a positive integer that is equal to the sum of its proper factors (its positive integer factors excluding the number itself). For example, 6 (with proper factors 1, 2 and 3) and 28 (with proper factors 1, 2, 4, 7 and 14) are perfect numbers.

   Write a program that prompts the user for a positive integer and finds all the perfect numbers up to and including that number.

   Save this program as `p16p4.py`.

Please upload your work to the Brightspace site before Sunday evening.

You should keep a copy of your programs for your portfolio.